



C++ Programming

Intern

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Project no. : Second

Project name: Login and Registration System

Login and Registration System

Introduction:

Login and Registration System is a simple command-line application developed using C++. The program allows users to register a new account, log in using their username and password, and search for their account when they forget their password..

This project was developed to practice fundamental C++ programming concepts, functions, file handling, input validation, and menu-driven programming.

Tools & Technologies Used:

- VS Code
- C++
- GitHub

Objectives of the Project:

The main objectives of the Login and Registration System are:

- To develop a simple login and registration application using C++.
- To practice functions and function prototypes.
- To understand and implement file handling.
- To learn how to read data from a text file.
- To learn how to write and append data to a text file.
- To practice loops and conditional statements.
- To use switch statements for menu selection.
- To take and process user input.
- To validate user registration information.
- To prevent duplicate usernames.
- To verify login credentials.
- To provide a forgot-password feature.
- To gain hands-on experience in developing a command-line application.

Features:

Following are the features of the Login and Registration System:

1. Displays a welcome screen and main menu.
2. Allows users to log in using a username and password.
3. Allows new users to register an account.
4. Stores registered usernames and passwords in "records.txt".
5. Checks whether a username already exists during registration.
6. Prevents duplicate usernames.

7. Checks username and password during login.
8. Displays a successful login message when credentials match.
9. Displays an error message when login credentials are incorrect.
10. Provides a forgot-password option.
11. Allows users to search for their account using their username.
12. Displays the stored password when the account is found.
13. Allows the user to return to the main menu.
14. Provides an Exit option.
15. Provides a simple and easy-to-use command-line interface.

C++ Concepts Used:

Following fundamental C++ Concepts are used in this project:

- Header files and Standard namespace

```
#include<iostream>
#include<fstream>
#include<string>
using namespace std;
```

- Functions

```
void login();
void registration();
void forgot();
```

- "if-else" statements

```
if(count==1)
{
    cout<<userId<<"\nYour Login is Successful!"<<endl;
    main();
}
else
{
    cout<<"Login Error! Please check your username and password."<<endl;
    cout<<endl;
    main();
}
```

- "switch" statements

```
switch (choice)
{
    case 1:
        login();
        break;
    case 2:
        registration();
        break;
    case 3:
        forgot();
        break;
    case 4:
        cout<<"Thank you for using the Login and Registration System. Goodbye!"<<endl;
        break;
    default:
        cout << "Invalid choice! Enter the valid Option given above." << endl;
}
```

- "while" loops

```
while(input>>id>>pass)
{
    if(id==userId && pass==password)
    {
        count=1;
    }
}
```

- String comparison

```
if(r_userId.empty() || r_password.empty())
{
    cout<<"Username and Password cannot be empty!"<<endl;
    main();
    return;
}
```

- "ifstream"

```
ifstream checkFile("records.txt");
string existingId, existingPass;
```

- "ofstream" and "ios::app"

```
ofstream f1("records.txt", ios::app);|
f1<<r_userId<<' '<<r_password<<endl;
```

Working of the Program:

The execution process of the program is as follows:

1. Three functions, "login()", "registration()", and "forgot()" are declared before the "main()" function.

```
void login();
void registration();
void forgot();
```

2. The program starts by displaying the title "Login and Registration System" and a welcome message. The main menu displays four options: Login, Register, Forgot Password and Exit. The user enters a choice from the main menu.

```
int main()
{
    cout<<"===== "<<endl;
    cout<<"          WELCOME          "<<endl;
    cout<<"    Login and Registration System    "<<endl;
    cout<<"===== "<<endl;
    cout<<endl;

    int choice;
    cout<<"----MENU----"<<endl;
    cout<<"1. Login" << endl;
    cout<<"2. Register" << endl;
    cout<<"3. Forgot Password" << endl;
    cout<<"4. Exit" << endl;
    cout<<endl;
    cout<<"Enter your choice: ";
    cin >> choice;
```

3. A "switch" statement checks the user's choice and calls the corresponding function.

```

switch (choice)
{
    case 1:
        login();
        break;
    case 2:
        registration();
        break;
    case 3:
        forgot();
        break;
    case 4:
        cout<<"Thank you for using the Login and Registration System. Goodbye!"<<endl;
        break;
    default:
        cout << "Invalid choice! Enter the valid Option given above." << endl;
}

cout<<endl;
return 0;

```

4. If the user selects Login, the "login()" function asks for the username and password. The program opens "records.txt" using "ifstream" and reads the stored usernames and passwords. A "while" loop compares the entered username and password with the records stored in the file. If both username and password match, the program displays a successful login message. If the credentials do not match, the program displays a login error message.

```

//Login function
void login()
{
    int count=0;
    string userId, password, id, pass;
    cout<<"-----Enter the Login Details-----"<<endl;
    cout<<"Enter the Username: ";
    cin>>userId;
    cout<<"Enter the Password: ";
    cin>>password;

    ifstream input("records.txt");

    while(input>>id>>pass)
    {
        if(id==userId && pass==password)
        {
            count=1;
        }
    }
    input.close();

    if(count==1)
    {
        cout<<userId<<"\nYour Login is Successful!"<<endl;
        main();
    }
    else
    {
        cout<<"Login Error! Please check your username and password."<<endl;
        cout<<endl;
        main();
    }
}

```

5. If the user selects Register, the "registration()" function asks the user to enter a username and password. The program checks whether the username or password is empty. The program then opens "records.txt" and checks whether the entered username already exists. If the username already exists, the program displays an error message and asks the user to choose another username.

```

// Registration function
void registration()
{
    string r_userId, r_password, r_id, r_pass;

    cout<<"-----Enter the Registration Details-----"<<endl;
    cout<<"Enter the Username: ";
    cin>>r_userId;
    cout<<"Enter the Password: ";
    cin>>r_password;

    if(r_userId.empty() || r_password.empty())
    {
        cout<<"Username and Password cannot be empty!"<<endl;
        main();
        return;
    }

    ifstream checkFile("records.txt");
    string existingId, existingPass;
    while(checkFile >> existingId >> existingPass)
    {
        if(existingId == r_userId)
        {
            cout<<"Username already exists! Please choose another username."<<endl<<endl;
            checkFile.close();
            main();
            return;
        }
    }
    checkFile.close();

    ofstream fi("records.txt", ios::app);
    fi<<r_userId<<" "<<r_password<<endl;
    fi.close();
    cout<<"Registration is Successful!"<<endl<<endl;
    main();
}

```

6. If the user selects Forget, the "forgot()" function asks the user to enter a username. When the user enters a username, the program searches "records.txt" for the matching account. If the username is found, the program displays the stored password.

```
// forgot function in case user forgets the password.
void forgot()
{
    int option;

    cout<<"-----Forgot Password-----"<<endl;
    cout<<"1. Search your account by Username"<<endl;
    cout<<"2. Go back to main menu"<<endl;
    cout<<endl;

    cout<<"Enter your choice: ";
    cin>>option;

    switch(option)
    {
        case 1:
        {
            int count=0;
            string s_userId, s_id, s_pass;
            cout<<"Enter the Username you remembered: ";
            cin>>s_userId;

            ifstream f2("records.txt");
            while(f2>>s_id>>s_pass)
            {
                if(s_id==s_userId)
                {
                    count=1;
                    break;
                }
            }
            f2.close();

            if(count==1)
            {
                cout<<"\nYour account is found!"<<endl;
                cout<<"\nYour password is: "<<s_pass<<endl;
            }
        }
    }
}
```

```
ifstream f2("records.txt");
while(f2>>s_id>>s_pass)
{
    if(s_id==s_userId)
    {
        count=1;
        break;
    }
}
f2.close();

if(count==1)
{
    cout<<"\nYour account is found!"<<endl;
    cout<<"\nYour password is: "<<s_pass<<endl;
    main();
}
else
{
    cout<<"\nSorry! Your account is not found!"<<endl;
    main();
}
break;

case 2:
{
    main();
}
break;

default:
cout<<"Invalid choice! Please try again."<<endl;
forgot();
}
```

Application Flow:

The overall basic workflow of the application is as follow:

Start → Display Welcome Screen → Display Main Menu → Select Option → Login / Register / Forgot Password / Exit → Process User Request → Display Result → Return to Main Menu → END

Code:

// LOGIN and REGISTRATION SYSTEM

#include<iostream>

#include<fstream>

#include<string>

using namespace std;

void login();

void registration();

void forgot();

int main()

{

cout<<"===== "<<endl;

cout<<" WELCOME "<<endl;

```

cout<<" Login and Registration System "<<endl;
cout<<"===== "<<endl;
cout<<endl;
int choice;
cout<<"----MENU----"<<endl;
cout<<"1. Login" << endl;
cout<<"2. Register" << endl;
cout<<"3. Forgot Password" << endl;
cout<<"4. Exit" << endl;
cout<<endl;
cout<<"Enter your choice: ";
cin >> choice;
switch (choice)
{
    case 1:
        login();
        break;
    case 2:
        registration();
        break;
    case 3:
        forgot();
        break;
    case 4:
        cout<<"Thank you for using the Login and Registration System.
Goodbye!"<<endl;
        break;
    default:
        cout << "Invalid choice! Enter the valid Option given above." << endl;
}
cout<<endl;

```

```

    return 0;
}
//Login function
void login()
{
    int count=0;
    string userId, password, id, pass;
    cout<<"-----Enter the Login Details-----"<<endl;
    cout<<"Enter the Username: ";
    cin>>userId;
    cout<<"Enter the Password: ";
    cin>>password;
    ifstream input("records.txt");
    while(input>>id>>pass)
    {
        if(id==userId && pass==password)
        {
            count=1;
        }
    }
    input.close();

    if(count==1)
    {
        cout<<userId<<"\nYour Login is Successful!"<<endl;
        main();
    }
    else
    {
        cout<<"Login Error! Please check your username and password."<<endl;
        cout<<endl;
    }
}

```

```

    main();
}
}
// Registration function
void registration()
{
    string r_userId, r_password, r_id, r_pass;
    cout<<"-----Enter the Registration Details-----"<<endl;
    cout<<"Enter the Username: ";
    cin>>r_userId;
    cout<<"Enter the Password: ";
    cin>>r_password;
    if(r_userId.empty() || r_password.empty())
    {
        cout<<"Username and Password cannot be empty!"<<endl;
        main();
        return;
    }

    ifstream checkFile("records.txt");
    string existingId, existingPass;
    while(checkFile >> existingId >> existingPass)
    {
        if(existingId == r_userId)
        {
            cout<<"Username    already    exists!    Please    choose    another
username."<<endl<<endl;
            checkFile.close();
            main();
            return;
        }
    }
}

```

```

}
checkFile.close();

ofstream f1("records.txt", ios::app);
f1<<r_userId<<' '<<r_password<<endl;
f1.close();
cout<<"Registration is Successful!"<<endl<<endl;
main();
}
// Forgot Function in case user forgets the password.
void forgot()
{
    int option;
    cout<<"-----Forgot Password-----"<<endl;
    cout<<"1. Search your account by Username"<<endl;
    cout<<"2. Go back to main menu"<<endl;
    cout<<endl;

    cout<<"Enter your choice: ";
    cin>>option;
    switch(option)
    {
        case 1:
        {
            int count=0;
            string s_userId, s_id, s_pass;
            cout<<"Enter the Username you remembered: ";
            cin>>s_userId;

            ifstream f2("records.txt");
            while(f2>>s_id>>s_pass)

```

```

{
    if(s_id==s_userId)
    {
        count=1;
        break;
    }
}
f2.close();
if(count==1)
{
    cout<<"\nYour account is found!"<<endl;
    cout<<"\nYour password is: "<<s_pass<<endl;
    main();
}
else
{
    cout<<"\nSorry! Your account is not found!"<<endl;
    main();
}
}
break;
case 2:
{
    main();
}
break;
default:
    cout<<"Invalid choice! Please try again."<<endl;
    forgot();
}
}

```

Sample Output:

```
PS C:\Users\amina\OneDrive\Desktop\VS CODE folder> cd "c:\Users\amin
ogin }
```

```
=====
WELCOME
Login and Registration System
=====
```

```
---MENU---
```

```
1. Login
2. Register
3. Forgot Password
4. Exit
```

```
Enter your choice: 2
----Enter the Registration Details----
```

```
Enter the Username: Ahmed
Enter the Password: 7654hg
Registration is Successful!
```

```
=====
WELCOME
Login and Registration System
=====
```

```
---MENU---
```

```
1. Login
2. Register
3. Forgot Password
4. Exit
```

```
Enter your choice: 1
----Enter the Login Details----
```

```
Enter the Username: Ahmed
Enter the Password: 7654hg
Ahmed
Your Login is Successful!
```

```
=====
WELCOME
Login and Registration System
=====
```

```
---MENU---
```

```
1. Login
2. Register
3. Forgot Password
4. Exit
```

```
Enter your choice: 4
Thank you for using the Login and Registration System. Goodbye!
```

```
Your Login is Successful!
=====
WELCOME
Login and Registration System
=====
```

```
---MENU---
```

```
1. Login
2. Register
3. Forgot Password
4. Exit
```

```
Enter your choice: 3
----Forgot Password-----
1. Search your account by Username
2. Go back to main menu
```

```
Enter your choice: 1
Enter the Username you remembered: Ahmed
```

```
Your account is found!

Your password is: 7654hg
=====
WELCOME
Login and Registration System
=====
```

Testing:

The application was tested by performing each operation:

Sr.	Task	Result	Status
1.	Main menu	Login, Register, Forgot Password, and Exit options are displayed correctly.	Passed
2.	User registration	Username and password are accepted and stored correctly	Passed
3.	User Login	Correct username and password result in successful login.	Passed
4.	Forgot password	Existing account can be searched using username.	Passed
5.	Exit option	Program exits with a goodbye message.	Passed

Conclusion:

The Login and Registration System successfully demonstrates the practical use of fundamental C++ programming concepts. The project helped in understanding functions, file handling, strings, loops, conditional statements, switch statements, user input, and menu-driven programming. The project also provided practical experience in storing and retrieving user information using a text file.

GitHub Repository:

[AmimaBatool/CodeAlpha Login-Registration-System: A C++ command-line application that allows users to register, log in, and recover forgotten passwords using file handling.](#)

Project status: Completed.